

Customer Churn Analysis

Summary

- Performed data cleaning by handling missing values in the **TotalCharges** column and converting data types for accurate analysis.
- Analyzed customer churn distribution and found that approximately **26.54% customers churned** from the telecom service.
- Identified that **senior citizens show comparatively higher churn behavior** than younger customers.
- Observed that customers with **short tenure (1–2 months)** are more likely to churn, while long-term customers tend to stay loyal.
- Found that customers with **month-to-month contracts** have the highest churn rate compared to yearly or two-year contracts.
- Service-based analysis revealed that customers **without OnlineSecurity and TechSupport services**, especially **Fiber optic internet users**, experience higher churn.
- Payment analysis showed that customers using the **Electronic Check payment method** have the highest churn tendency.

Overall, the analysis indicates that contract type, tenure, internet service type, additional support services, and payment methods are the major factors influencing customer churn.

Recommendations

- Encourage customers to switch from **month-to-month contracts** to long-term plans by offering discounts and loyalty benefits.
- Improve customer retention by providing bundled services such as **OnlineSecurity** and **TechSupport** at affordable prices.
- Focus on improving the experience of **Fiber optic internet users**, as they contribute significantly to churn.

- Create targeted retention campaigns for **new customers** during their initial months of service.
- Offer incentives for customers using **Electronic Check payments** to migrate to more stable payment methods like auto-pay or credit card billing.
- Develop personalized support and engagement strategies for **senior citizen customers** to reduce churn risk.